Damandeep Singh

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Summary

Machine learning engineer and M.S. Computer Science student at Texas A&M with 3 years building production ML and LLM systems. Experience across LLM agents, transformers, deep learning, computer vision, federated learning, and meta-learning, with GPU training on NVIDIA A100s and end-to-end ownership of high-scale backend services.

Education

Texas A&M University — College Station, TX
Master of Science in Computer Science — *Aug 2026 – Dec 2027 (Expected)*
- *Relevant Coursework:* Artificial Intelligence, Natural Language Processing (LLMs), Analysis of Algorithms

National Institute of Technology Srinagar — Srinagar, India
Bachelor of Technology in Information Technology; CGPA: 8.30/10 (Top 10%) — *Jul 2019 – Jun 2023*

Experience

Software Engineer — Jul 2024 – Jun 2026
MediBuddy — *Bengaluru, India*
- Built a natural-language-to-SQL system with LLM agents over Model Context Protocol (MCP) that introspect schemas, share context, and generate SQL; used by operations to report metrics and catch order mismatches without engineers.
- Trained custom document ML models (Amazon Textract adapters) on historical prescriptions to extract key fields and classify documents as valid or invalid, replacing a 10-minute manual review with automated decisions.
- Owned the Meds service end to end (TypeScript, NestJS, PostgreSQL, AWS), one of MediBuddy's three core revenue streams handling 15K+ orders/day (450K+/month), including design, delivery, and on-call support.
- Built billing reconciliation with FIFO queue-based idempotency and state machines, cutting monthly financial leakage by 89%.
- Designed vendor assignment via weighted scoring and a greedy heuristic with sub-100ms Redis-cached lookups; cut delivery time from 48 to under 6 hours.

Software Engineer — Sep 2023 – Jul 2024
ShipThis — *Bengaluru, India*
- Built a document-parsing pipeline with Azure Document Intelligence and custom OCR models extracting structured data from emails, processing 1,000+ documents/day.

Research Experience

Graduate Student Researcher — Aug 2026
Texas A&M University — *College Station, TX*
- Worked on long event-sequence modeling with a causal temporal transformer, forecasting daily value trajectories that feed a downstream 30-day risk prediction model; set up training on NVIDIA A100 GPUs (TAMU HPRC).

Machine Learning Intern — Dec 2021 – Mar 2022
IIT Hyderabad — *Hyderabad, India*
- Built real-time facial recognition with MediaPipe FaceMesh (468 landmarks); cut inference latency by 20%.

Machine Learning Intern — Aug 2021 – Oct 2021
IIT Delhi — *New Delhi, India*
- Implemented First-Order MAML for few-shot N-way K-shot classification; benchmarked convergence against Reptile.

Projects

Federated Learning for Fraud Detection | *Python, PyTorch, FedAvg*
- Implemented FedAvg in PyTorch across 12 to 32 simulated clients with partial participation per round, keeping raw data local; compared imbalanced vs. SMOTE-balanced training on minority-class precision, recall, and F1.

Sentinel: MCP Security Gateway for AI Agents | *Go, OpenFGA, JSON-RPC* | github.com/Damandeep02/sentinel-mcp-gateway
- Built a Go gateway intercepting MCP JSON-RPC traffic (stdio, HTTP/SSE) to enforce per-agent authorization (rules or OpenFGA), with fail-closed intent validation for high-risk tool calls and PII redaction.

LLM Agent for Trustworthy Image-Based Assessment | *Python, LLMs, Multimodal* — Oct 2026 – Present
- Developing an LLM agent that grounds its assessments of ultrasound images in verifiable evidence; co-authored project proposal (NLP graduate course).

Technical Skills

ML / AI: Machine Learning, Deep Learning, LLMs, Transformers, AI Agents, RAG, NLP, Computer Vision, Federated Learning
ML Libraries: PyTorch, TensorFlow, scikit-learn, LangChain, OpenCV, MediaPipe, NumPy, Pandas
Languages: Python, C++, Go, TypeScript, JavaScript, Java, SQL
Systems & Cloud: GPU Training (NVIDIA A100), AWS, Azure, Docker, PostgreSQL, Redis, Kafka, Git, CI/CD

Honors & Awards

- **2nd Place, MediBuddy AI/ML Hackathon:** Built Doctor Copilot, an LLM-based decision-support proof of concept.
- **Winner, Devfolio Base Hackathon (Oct 2024):** 1 of 16 winning teams across 4 countries.